47 Chapter Review

Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

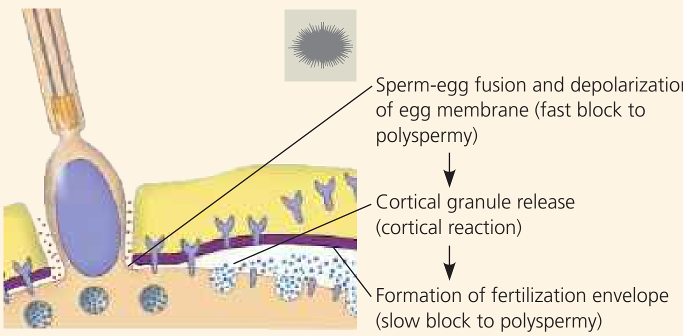
SUMMARY OF KEY CONCEPTS

To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkjz9t.

CONCEPT **47.1**

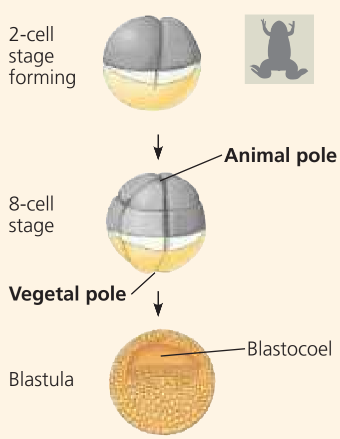
Fertilization and cleavage initiate embryonic development *(pp. 1044–1049)*

- **Fertilization** forms a diploid zygote and initiates embryonic development. The **acrosomal reaction** releases hydrolytic enzymes from the sperm head that digest material surrounding the egg.



In mammalian fertilization, the cortical reaction modifies the **zona pellucida** as a slow block to polyspermy.

- Fertilization is followed by **cleavage**, a period of rapid cell division without growth, producing a large number of cells called **blastomeres**. The amount and distribution of **yolk** strongly influence the pattern of cleavage. In many species, the completion of the cleavage stage generates a **blastula** containing a fluid-filled cavity, the **blastocoel**.



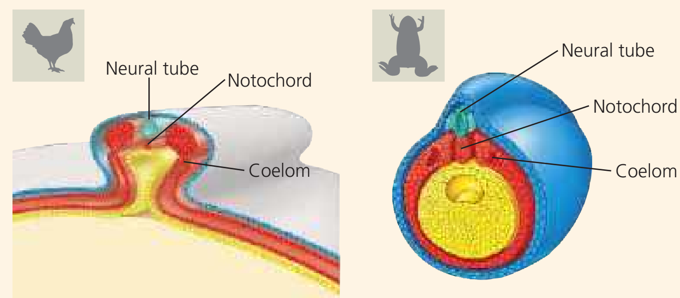
What cell-surface event would likely fail if a sperm contacted an egg of another species?

CONCEPT **47.2**

Morphogenesis in animals involves specific changes in cell shape, position, and survival *(pp. 1049–1057)*

- **Gastrulation** converts the blastula to a **gastrula**, which has a primitive digestive cavity and three **germ layers: ectoderm** (blue), which forms the outer layer of the embryo, **mesoderm** (red), which forms the middle layer, and **endoderm** (yellow), which gives rise to the innermost tissues.

- Gastrulation and organogenesis in mammals resemble the processes in reptiles, including birds. After fertilization and early cleavage in the oviduct, the **blastocyst** implants in the uterus. The **trophoblast** initiates formation of the fetal portion of the placenta, and the embryo proper develops from a cell layer, the epiblast, within the blastocyst.
- The embryos of birds, other reptiles, and mammals develop within a fluid-filled sac that is contained within a shell or the uterus. In these organisms, the three germ layers produce four **extraembryonic membranes**: the amnion, chorion, yolk sac, and allantois.
- The organs of the animal body develop from specific portions of the three embryonic germ layers. Early events in **organogenesis** in vertebrates include neurulation: formation of the **notochord** by cells of the dorsal mesoderm and development of the **neural tube** from infolding of the ectodermal neural plate.



- Cytoskeletal rearrangements cause changes in the shape of cells that underlie cell movements in gastrulation and organogenesis, including invaginations and **convergent extension**. The cytoskeleton is also involved in cell migration, which relies on cell adhesion molecules and the extracellular matrix to help cells reach specific destinations. Migratory cells arise both from the neural crest and from **somites**.
- Some processes in animal development require **apoptosis**, programmed cell death.

What are some functions of apoptosis in development?

CONCEPT **47.3**

Cytoplasmic determinants and inductive signals regulate cell fate *(pp. 1057–1064)*

- Experimentally derived **fate maps** of embryos show that specific regions of the zygote or blastula develop into specific parts of older embryos. The complete cell lineage has been worked out for *C. elegans*, revealing that programmed cell death contributes to animal development. In all species, the developmental potential of cells becomes progressively more limited as embryonic development proceeds.
- Cells in a developing embryo receive and respond to **positional information** that varies with location. This information is often in the form of signaling molecules secreted by cells in specific regions of the embryo, such as the dorsal lip of the blastopore in the amphibian gastrula and the **apical ectodermal ridge** and **zone of polarizing activity** of the vertebrate limb bud.

Suppose you found two classes of mouse mutations, one that affected limb development only and one that affected both limb and kidney development. Which class would be more likely to alter the function of monocilia? Explain.